$$\Rightarrow \qquad 1 = \frac{200\sqrt{3}}{BD} \quad [\text{by } (1)]$$

we get, $BD = 200\sqrt{3}$

Now, $CD = BD - BC$

$$CD = 200\sqrt{3} - 200 = 200(\sqrt{3} - 1) = 146.4$$

It is given that the distance CD is covered in 10 seconds.

That is, the distance of 146.4 m is covered in 10 seconds.

Therefore, speed of the boat $= \dfrac{\text{distance}}{\text{time}}$

$$= \frac{146.4}{10} = 14.64 \text{ m/s} \Rightarrow 14.64 \times \frac{3600}{1000} \text{ km/hr} = 52.704 \text{ km/hr}$$

Exercise 6.3

1. From the top of a rock $50\sqrt{3}$ m high, the angle of depression of a car on the ground is observed to be 30°. Find the distance of the car from the rock.
2. The horizontal distance between two buildings is 70 m. The angle of depression of the top of the first building when seen from the top of the second building is $45°$. If the height of the second building is 120 m, find the height of the first building.
3. From the top of the tower 60 m high the angles of depression of the top and bottom of a vertical lamp post are observed to be $38°$ and $60°$ respectively. Find the height of the lamp post. $(\tan 38° = 0.7813, \ \sqrt{3} = 1.732)$
4. An aeroplane at an altitude of 1800 m finds that two boats are sailing towards it in the same direction. The angles of depression of the boats as observed from the aeroplane are $60°$ and $30°$ respectively. Find the distance between the two boats. $(\sqrt{3} = 1.732)$
5. From the top of a lighthouse, the angle of depression of two ships on the opposite sides of it are observed to be $30°$ and $60°$. If the height of the lighthouse is h meters and the line joining the ships passes through the foot of the lighthouse, show that the distance between the ships is $\dfrac{4h}{\sqrt{3}}$ m.
6. A lift in a building of height 90 feet with transparent glass walls is descending from the top of the building. At the top of the building, the angle of depression to a fountain in the garden is $60°$. Two minutes later, the angle of depression reduces to $30°$. If the fountain is $30\sqrt{3}$ feet from the entrance of the lift, find the speed of the lift which is descending.

6 Trigonometry.indd 261 15-12-2020 03:13:59 PM